

Abstract

Vehicular Ad Hoc Networks (VANETs) form the backbone of modern intelligent transportation systems, enabling real-time communication between vehicles and roadside infrastructure. However, their decentralized and highly mobile nature exposes them to numerous cyber-threats, including gray-hole attacks, spoofing, false message injection, jamming, and CAN-bus manipulation. Conventional machine-learning-based intrusion detection systems require centralized data aggregation, raising privacy concerns, scalability limitations, and high communication overhead. To address these challenges, this work proposes a Federated Learning-enabled Intrusion Detection and Prevention System (FL-IDPS) that collaboratively learns global threat patterns without sharing raw vehicular data.